JAMA | Review

Atrial Fibrillation
A Review

Darae Ko, MD, MSc; Mina K. Chung, MD; Peter T. Evans, MD; Emelia J. Benjamin, MD, ScM; Robert H. Helm, MD

IMPORTANCE In the US, approximately 10.55 million adults have atrial fibrillation (AF). AF is associated with significantly increased risk of stroke, heart failure, myocardial infarction, dementia, chronic kidney disease, and mortality.

OBSERVATIONS Symptoms of AF include palpitations, dyspnea, chest pain, presyncope, exertional intolerance, and fatigue, although approximately 10% to 40% of people with AF are asymptomatic. AF can be detected incidentally during clinical encounters, with wearable devices, or through interrogation of cardiac implanted electronic devices. In patients presenting with ischemic stroke without diagnosed AF, an implantable loop recorder (ie, subcutaneous telemetry device) can evaluate patients for intermittent AF. The 2023 American College of Cardiology (ACC)/American Heart Association (AHA)/American College of Clinical Pharmacy (ACCP)/Heart Rhythm Society (HRS) Guideline writing group proposed 4 stages of AF evolution: stage 1, at risk, defined as patients with AF-associated risk factors (eg, obesity, hypertension); stage 2, pre-AF, signs of atrial pathology on electrocardiogram or imaging without AF; stage 3, the presence of paroxysmal (recurrent AF episodes lasting ≤7 days) or persistent (continuous AF episode lasting >7 days) AF subtypes; and stage 4, permanent AF. Lifestyle and risk factor modification, including weight loss and exercise, to prevent AF onset, recurrence, and complications are recommended for all stages. In patients with estimated risk of stroke and thromboembolic events of 2% or greater per year, anticoagulation with a vitamin K antagonist or direct oral anticoagulant reduces stroke risk by 60% to 80% compared with placebo. In most patients, a direct oral anticoagulant, such as apixaban, rivaroxaban, or edoxaban, is recommended over warfarin because of lower bleeding risks. Compared with anticoagulation, aspirin is associated with poorer efficacy and is not recommended for stroke prevention. Early rhythm control with antiarrhythmic drugs or catheter ablation to restore and maintain sinus rhythm is recommended by the 2023 ACC/AHA/ACCP/HRS Guideline for some patients with AF. Catheter ablation is first-line therapy in patients with symptomatic paroxysmal AF to improve symptoms and slow progression to persistent AF. Catheter ablation is also recommended for patients with AF who have heart failure with reduced ejection fraction (HFrEF) to improve quality of life, left ventricular systolic function, and cardiovascular outcomes, such as rates of mortality and heart failure hospitalization.

CONCLUSIONS AND RELEVANCE AF is associated with increased rates of stroke, heart failure, and mortality. Lifestyle and risk factor modification are recommended to prevent AF onset, recurrence, and complications, and oral anticoagulants are recommended for those with an estimated risk of stroke or thromboembolic events of 2% or greater per year. Early rhythm control using antiarrhythmic drugs or catheter ablation is recommended in select patients with AF experiencing symptomatic paroxysmal AF or HFrEF.

JAMA. 2025;333(4):329-342. doi:10.1001/jama.2024.22451
Published online December 16, 2024.

Multimedia

CME at jamacmelookup.com

Author Affiliations: Author affiliations are listed at the end of this article.

Corresponding Author: Emelia J. Benjamin, MD, ScM, Boston University Chobanian and Avedisian School of Medicine, Cardiology, Collamore 8, 72 E Concord St, Boston, MA 02118 (emelia@bu.edu).

Section Editor: Kristin Walter, MD, Deputy Editor.

© 2024 American Medical Association. All rights reserved, including those for text and data mining, AI training, and similar technologies.

Downloaded from jamanetwork.com by Northern Arizona University user on 05/25/2025

In the US, atrial fibrillation (AF) affects up to 1 in 3 people in their lifetime and was estimated to affect approximately 10.55 million (95% CI, 10.48-10.62 million) people by 2019.[1-3] Significant complications associated with AF include ischemic stroke, heart failure (HF), myocardial infarction, chronic kidney disease, dementia, and mortality. Oral anticoagulants (OACs) have reduced rates of stroke and mortality in patients with AF.[4] This review summarizes current evidence regarding the epidemiology, pathophysiology, diagnosis, and management of AF (**Box**).

Methods

We conducted a PubMed search for English-language articles published between January 1, 1990, and August 15, 2024, about the epidemiology, pathophysiology, clinical presentation, prognosis, and management of AF. Current guidelines, randomized clinical trials, and studies with larger sample sizes were prioritized for inclusion. The 107 articles comprising this review included 48 randomized clinical trials; 19 meta-analyses; 12 guidelines, consensus documents, or scientific statements; 23 longitudinal and 2 cross-sectional observational studies; and 3 reviews.

Epidemiology

The incidence, prevalence, and lifetime risk of AF are increasing,[5,6] most likely due to population aging, increased detection rates, and increased survival with AF and other cardiovascular diseases.[6,7] Spanning 50 years of Framingham Heart Study participant surveillance (1958-1967 and 1998-2007), the age-adjusted incidence per 1000 person-years increased from 3.7 to 13.4 in men and 2.5 to 8.6 in women and the prevalence increased from 20.4 to 96.2 in men and 13.7 to 49.4 in women.[7]

The incidence and prevalence of AF vary by region and demographic factors, including sex and age. The Global Burden of Disease project estimated that in 2021, 52.55 million individuals (95% CI, 43.49-63.74 million) worldwide had AF or atrial flutter, with the highest prevalence observed in high-income countries in North America, Australasia, and Western Europe.[5] The global prevalence was higher in men (~28 million) vs women (~25 million).[5] Older age is associated with higher incidence of AF (hazard ratio [HR] per 5-year increase in age, 1.66 [95% CI, 1.59-1.74]).[8]

Older age, current smoking, taller height, greater weight, higher blood pressure (systolic, diastolic, and hypertension treatment), presence of diabetes, and presence of heart disease (HF or myocardial infarction) are associated with higher rates of AF.[8-10] Risk of AF is also affected by genetic factors. Compared with individuals in the upper third of both clinical (CHARGE-AF score) and polygenic risk (48.2% [95% CI, 41.3%-55.1%]), individuals in the lower third had less than half the lifetime risk of AF (22.3% [95% CI, 15.4%-29.1%]).[2]

Additional factors associated with increased AF risk include more than moderate alcohol use (defined as >1 standard alcoholic drink per day or binge drinking), sleep apnea, and hyperthyroidism.[4,11-13] Compared with sedentary lifestyle, physical activity is associated with lower risk of AF. However, male endurance athletes have higher risk.[4,14] Although most data relating risk factors to AF are observational, Mendelian randomization studies support causal relations between multiple risk factors, including adiposity, smoking, alcohol consumption, and hypertension.[4,15]

Box. Frequently Asked Questions About Atrial Fibrillation (AF)

Among patients with AF, who should receive anticoagulation for stroke and thromboembolism prevention?

In patients with estimated risk of ischemic stroke or thromboembolic events of 2% or greater per year, benefits of anticoagulation exceed risks of major bleeding. Several risk scores identify patients at high risk of stroke, including CHA_2DS_2-VASc score, which assigns 1 point for congestive heart failure, 1 point for hypertension, 1 point for age 65 years and older, 2 points for age 75 years and older, 2 points for previous stroke or transient ischemic attack, 1 point for vascular disease, and 1 point for female sex. Anticoagulation should be continued indefinitely unless a contraindication, such as bleeding, develops.

What therapies should be recommended for patients at increased risk of AF?

Lifestyle and risk factor modification, including weight loss, moderate exercise, smoking cessation, reducing alcohol consumption, and optimal blood pressure control, are recommended to prevent AF in patients at increased risk.

How should an asymptomatic patient with new-onset AF be treated?

Patients younger than 70 years with new-onset AF may benefit from rhythm control with cardioversion and initiation of an antiarrhythmic drug, even if AF is asymptomatic. Rhythm control should also be considered for patients with heart failure or left ventricular systolic dysfunction. Anticoagulation should be initiated if the stroke or thromboembolic event risk score is 2% or greater per year.

Pathophysiology

Ectopic atrial premature beats that initiate AF typically arise from myocardial cells (sleeves) that extend a few centimeters from the pulmonary vein–atrial junction into the pulmonary veins. Although atrial ectopy may initiate AF, persistence of AF is often due to disease-specific atrial electrophysiologic, structural, and histopathologic changes that promote electrical reentry and AF continuance.[16] For example, hypertension activates the renin-angiotensin-aldosterone system, which induces atrial fibrosis and hypertrophy, slowing atrial conduction and promoting reentry and AF.[17] Obesity increases oxidative stress, systemic inflammation, and abnormal Ca^{2+}cycling. These effects increase atrial ectopy and pathologic changes that sustain arrhythmic activity. Autonomic dysfunction (eg, with sleep-disordered breathing) alters atrial repolarization and promotes AF.[17] AF-associated diseases, such as hypertension, obesity, and valvular heart disease (eg, mitral valve stenosis, mitral valve regurgitation), are associated with atrial pathology and AF.[18]

AF Screening and Detection

The probability of AF detection increases with electrocardiogram (ECG) monitoring duration. The benefits of screening for AF in the general population to detect asymptomatic AF are currently uncertain.[4] An implantable loop recorder (ILR) is a subcutaneous device that can continuously monitor heart rhythm for about 4 years. The LOOP Study (Implantable Loop Recorder Detection of Atrial Fibrillation to Prevent Stroke) randomized 6004 patients aged 70 to 90 years with 1 or more of 4 comorbidities (hypertension, diabetes, prior stroke, or HF) to ILR or usual care.[19] During

© 2024 American Medical Association. All rights reserved, including those for text and data mining, AI training, and similar technologies.

Downloaded from jamanetwork.com by Northern Arizona University user on 05/25/2025

a median of 64.5 months of follow-up, AF was diagnosed in 31.8% of the ILR group vs 12.2% of the control group. Even though OACs were initiated in patients diagnosed with incident AF, there was no significant difference (HR, 0.80 [95% CI, 0.61-1.05]) in risk of the primary outcome of stroke or systemic arterial embolism in patients randomized to ILR (0.88 events per 100 person-years [95% CI, 0.68-1.12] vs the control group, 1.09 events [95% CI, 0.96-1.24]).

Many individuals use wearable devices that evaluate heart rhythm for AF. In the Apple Heart Study, only 34% of those who received the smartwatch's notification for AF were subsequently diagnosed with AF using ECG patch monitoring.[20] AF detected by consumer wearable devices (eg, smartwatches) should be confirmed with ECG correlation or through additional testing (eg, with an ECG patch monitor).

In patients with ischemic stroke or systemic thromboembolism without established AF, ILR is a reasonable diagnostic measure to increase probability of AF detection if maximum sensitivity is sought[4] to identify patients who may benefit from OACs for secondary stroke prevention. For secondary stroke prevention, OACs are contraindicated in patients without AF or in those without a known cardioembolic source.[21,22] In randomized clinical trials of patients with ischemic stroke, rates of AF detection were 12% to 15%[23-25] after 1 year and 30%[23] by 3 years with ILR, compared with rates of 1.8% to 4.7% after 1 year and 3% after 3 years in those who received usual care or wore a 30-day monitor.

AF diagnosed during hospitalization for noncardiac illnesses, such as sepsis or noncardiac surgery, is associated with posthospital AF recurrence, stroke, and mortality.[4,26,27] In patients with severe sepsis, new-onset AF was associated with 2.7-fold (2.6% vs 0.6%) increased risk of in-hospital ischemic stroke and 7% (56% vs 39%) higher risk of in-hospital death, adjusting for demographics, comorbidities, and sepsis-associated factors.[27] Patients diagnosed with AF during hospitalization for noncardiac illness have 5-year recurrence rates of 42% to 68%.[28,29] The 2023 American College of Cardiology (ACC)/American Heart Association (AHA)/American College of Clinical Pharmacy (ACCP)/Heart Rhythm Society (HRS) Guideline recommends that patients diagnosed with AF during hospitalization for noncardiac illness be counseled about their increased risk of recurrent AF.[4] Whether long-term OACs should be initiated at the time of hospital discharge or deferred until AF recurs during subsequent follow-up remains unclear.[30]

In patients without diagnosed AF who have cardiac implanted electronic devices, including an ILR, pacemaker, or implantable cardioverter-defibrillator, the incidence of atrial high-rate episodes (AHREs; defined as asymptomatic atrial tachyarrhythmias with atrial rates >190 beats per minute, which include AF, atrial flutter, and atrial tachycardias) detected is approximately 24.5% to 34.4% over 1 to 2.5 years of follow-up.[31,32] Such patients may or may not have underlying cardiomyopathy or conduction disease. Although the duration threshold for meaningful AHREs has varied among studies, from longer than 20 seconds to 24 hours, AHREs of greater than 6 minutes are associated with incident AF.[31] A meta-analysis of 15 353 participants reported that AHREs were associated with 2.4-fold increased risk of stroke (1.89 per 100 person-years with vs 0.93 per 100 person-years without) that increased with AHRE duration.[33] A meta-analysis of 2 randomized clinical trials that included 6548 patients studied whether OACs (vs no anticoagulation or low-dose aspirin) were associated with lower rates of stroke in those with AHRE lasting 6 minutes or longer. In this meta-analysis, direct OACs (DOACs) were associated with lower relative risk (RR) of stroke by 32% (2.0% vs 3.0%), but were associated with 39% increased RR of major bleeding (4.8% vs 3.2%).[34] The ischemic stroke rate was 1% per patient-year in the control group (patients with AHREs not on DOACs), a significant stroke risk with AHREs, but lower than the literature-estimated 2% per patient-year for patients with diagnosed AF (detected on 12-lead ECG). It is reasonable to initiate OACs for AHREs lasting longer than 24 hours in individuals with CHA_2DS_2-VASc scores of 2 or higher. It may be reasonable for AHREs lasting longer than 5 minutes in those with CHA_2DS_2-VASc score of 3 or higher, but bleeding risk should be carefully considered.[4]

Clinical Presentation

Typical symptoms of AF include palpitations with or without dyspnea, chest pain, presyncope, exertional intolerance, and fatigue. Approximately 10% to 40% of people with AF are asymptomatic.[4] Asymptomatic AF may be detected during routine clinical encounters or with a wearable monitor or cardiac implanted electronic device interrogation. Asymptomatic initial presentation of AF is more common in men (10% in men vs 3% in women) and older adults (mean age, 74 years vs 62 years for symptomatic people).[35,36] Diabetes is more common in those with asymptomatic AF. Asymptomatic AF may also be discovered during evaluation of AF-related clinical outcomes, such as ischemic stroke, systemic thromboembolism, myocardial infarction, or HF.

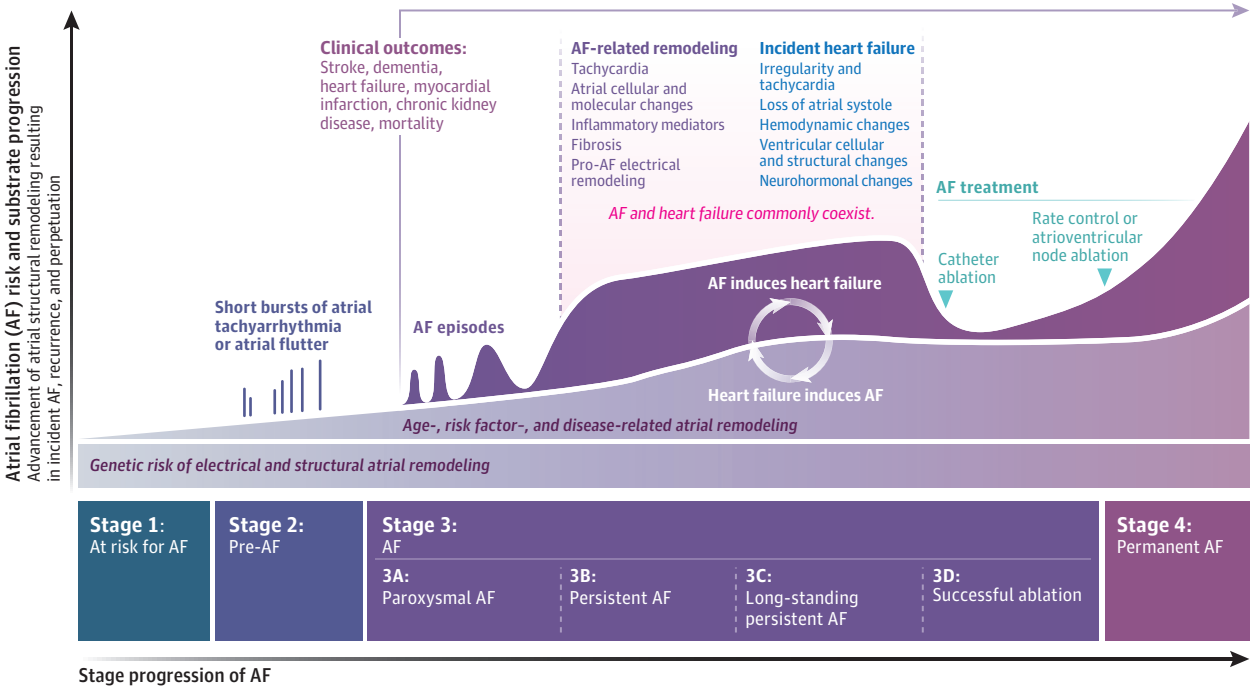
AF and HF are predisposing conditions for each other and often coexist at the time of incident AF (**Figure 1**).[37] In patients with AF and newly diagnosed HF and reduced ejection fraction (HFrEF), tachycardia-mediated cardiomyopathy should be considered when common etiologies (eg, ischemia) have been excluded.[4]

The diagnosis of AF is confirmed by identifying irregular atrial activity (fibrillatory waves) without discrete P waves on 12-lead ECG[4] or lasting for longer than 30 seconds on rhythm strip. AF is classified as paroxysmal (intermittent AF episodes lasting ≤7 days), persistent (continuous AF episodes lasting >7 days and/or requiring cardioversion), or long-standing persistent (AF episode lasting >1 year). AF subtype is determined by initial clinical presentation and may reflect severity of atrial pathology.[18] In some cases, the AF subtype may not be apparent at time of diagnosis; its persistence or spontaneous termination during the subsequent week allow for classification.

Evaluation of patients with newly identified AF should include transthoracic echocardiogram to assess cardiac structure and to identify possible causes for AF (eg, valve disease)[4] or outcomes associated with AF (eg, reduced ventricular function). Basic laboratory testing, including complete blood count, metabolic panel, and thyroid function, is appropriate.[4] Additional testing, such as cardiac stress testing in patients with newly diagnosed AF, should not be routinely performed without specific indications, such as angina or reduced ejection fraction.[4] Outpatient evaluation and management is appropriate for patients with asymptomatic or mildly symptomatic AF in the absence of HF, ischemia, or poorly controlled ventricular rates.

© 2024 American Medical Association. All rights reserved, including those for text and data mining, AI training, and similar technologies.
Downloaded from jamanetwork.com by Northern Arizona University user on 05/25/2025

Figure 1. Atrial Fibrillation Stages and the Evolution of Atrial Pathology



AF stages are reflective of the evolution of AF and include those at risk (stage 1), those with pre-AF (stage 2), those with AF (stage 3), and those with permanent AF (stage 4). Genetic-, age-, risk factor–, and disease-related remodeling generate the AF substrate, defined as any combination of molecular, cellular, structural, or electrical changes to the atrium that increase susceptibility for and perpetuation of AF. As AF substrate advances, AF may become more persistent. Heart failure and AF commonly coexist, and this figure highlights the feedforward interaction and effect of treatment on clinical outcomes.

Emergency department referral may be indicated for hemodynamically unstable or highly symptomatic AF, AF with concomitant HF, or AF with rapid ventricular rates. Emergent cardioversion should be considered for hemodynamic instability attributable to AF.[4] The risk of thromboembolism must be carefully considered, particularly if a patient has not been on therapeutic anticoagulation or if the AF duration has persisted for longer than 48 hours and imaging with transesophageal echocardiogram or cardiac computed tomography to exclude left atrial thrombus is not immediately feasible.

Classification Scheme: Stages of AF

AF is no longer categorized according to valvular or nonvalvular AF.[4] This distinction is only currently used to guide OAC strategy.

Four AF stages have been proposed (Figure 1).[4] Individuals with modifiable and nonmodifiable risk factors, such as obesity or family history of AF, are classified as stage 1 at risk for AF. Stage 2, pre-AF, is defined as the presence of atrial pathology, including left atrial enlargement, frequent atrial ectopy, or nonsustained atrial tachycardia, but without diagnosed AF. Individuals with conditions associated with high incidence of AF, such as atrial flutter, HF, coronary artery disease, valvular heart disease, hypertrophic cardiomyopathy, neuromuscular diseases, and hyperthyroidism, are considered to have pre-AF; clinicians may consider increased AF surveillance consisting of a wearable or implantable monitor. However, no randomized clinical trials have demonstrated that surveillance for AF prevents adverse outcomes. Stage 3 AF is clinically apparent AF and is categorized as 1 of 4 subtypes: 3a, paroxysmal (recurrent AF episodes lasting ≤7 days); 3C, persistent AF (continuous AF episode lasting >7 days); 3C, long-standing persistent AF (continuous AF episode lasting >1 year); or a new subtype, 3d, in which AF was treated successfully with catheter ablation. After catheter ablation, AF episodes may be less symptomatic, less frequent, or shorter in duration.[4] Individuals with stage 4 AF have permanent AF, for which a decision has been made not to pursue rhythm control based on patient and clinical factors, such as age and AF duration.

Prognosis

In a meta-analysis of 9 686 513 patients, compared with absence of diagnosed AF, AF was associated with absolute risk increase per 1000 participant-years of approximately 3.6 for stroke, 11.1 for HF, 1.4 for ischemic heart disease, 6.6 for chronic kidney disease, and 3.8 for mortality (**Table 1** includes RRs).[38] A meta-analysis reported AF was associated with increased risk of Alzheimer disease (adjusted OR [aOR], 1.4) and vascular dementia (aOR, 1.7), and in stroke-free patients (n = 324 494), cognitive impairment or dementia (adjusted HR [aHR], 1.4; absolute numbers unavailable).[40] In randomized clinical trials evaluating DOACs vs warfarin, patients who received apixaban (HR, 0.79) had an annual stroke or systemic embolism rate of 1.3% compared with 1.6% for warfarin.[41-44]

In a community-based cohort of 3491 adults with AF, incidence rates per 100 person-years of follow-up were 2.13 (95% CI, 1.88-2.40) for HFrEF and 3.32 (95% CI, 3.01-3.66) for HF with preserved ejection fraction (HFpEF).[45] Although AF is associated with increased rates of HF (Table 1), randomized clinical trials have not established treatments among people with AF that prevent HF.[6,38,39]

© 2024 American Medical Association. All rights reserved, including those for text and data mining, AI training, and similar technologies.
Downloaded from jamanetwork.com by Northern Arizona University user on 05/25/2025

Table 1. Relative, 5-Year, and Lifetime Risks and Restricted Mean Time Lost With Various Atrial Fibrillation (AF) Outcomes

	Odutayo et al[38]		Piccini et al[39]	Vinter et al[6]	
Study design	Systematic review and meta-analysis		Fee-for-service Medicare beneficiaries aged ≥65 y; incident AF, 1999-2007	Danish population-based study, 3.5 million individuals free of AF at index age 45 y, 2000-2022	
Outcome	Relative risk (95% CI)	Absolute rate per 1000 participant-y[a]	5-y risk, %	Lifetime risk, % (95% CI)[b]	Restricted mean time lost, y[c]
Stroke	2.42 (2.17-2.71)	3.6	7.1	21.4 (20.6-22.3)	6.2 (5.9-6.6)
Heart failure	4.99 (3.04-8.22)	11.1	13.7	41.2 (39.8-42.7)	13.6 (12.9-14.3)
Myocardial infarction	1.61 (1.38-1.87)[d]	1.4[d]	3.9	11.5 (10.9-12.2)	3.6 (3.3-3.9)
All-cause mortality	1.46 (1.39-1.53)	3.8	48.8		

[a] Absolute rate data are unavailable for Piccini et al[39] and Vinter et al.[6]

[b] Lifetime risk is the cumulative incidence function at age 95 years.

[c] Restricted mean time lost is the number of disease-free years lost by age 95 years.

[d] Odutayo et al meta-analyzed ischemic heart disease instead of myocardial infarction.

In a study of 9769 patients with AF who were taking 1 of 4 OACs (apixaban, dabigatran, rivaroxaban, or warfarin), use of DOACs was associated with lower rates of acute kidney injury (HR, 0.68 [95% CI, 0.58-0.81]).[9]

Treatments

Recommended treatment of patients at risk for AF (stages 1 or 2) or with AF (stages 3 or 4) consists of lifestyle and risk factor modification, such as weight loss, exercise, and targeted blood pressure control. However, except for hypertension treatment, randomized clinical trial evidence that these lifestyle and risk factors prevent AF does not exist. Lifestyle and risk factor modification is also recommended for patients with AF who are treated with antiarrhythmic drugs (AADs) or ablation (**Figure 2**).[4]

Primary Prevention: Risk Factors

The 2023 ACC/AHA/ACCP/HRS Guideline recommends lifestyle and risk factor modification for individuals with stages 1 and 2 AF, including treating obesity, diabetes, cigarette smoking, and hypertension, and recommendations to address physical inactivity and unhealthy alcohol consumption.[4] In secondary analysis of randomized clinical trial data, intensive blood pressure control (ie, lowering systolic blood pressure to <120 mm Hg, compared with <140 mm Hg) was associated with lower AF risk (6.21 vs 8.33 events per 1000 person-years; HR, 0.74 [95% CI, 0.56-0.98]).[46] A meta-analysis of 20 randomized clinical trials that included 63 604 patients with diabetes, HF, or kidney disease reported that sodium-glucose cotransporter-2 inhibitors were associated with reduced AF risk (RR, 0.82 [95% CI, 0.72-0.93]; absolute numbers unavailable).[47]

Secondary Prevention

Although treatment of lifestyle and risk factors has been recommended for many years, these preventive strategies are not widely implemented.[48-51] The 2023 ACC/AHA/ACCP/HRS Guideline emphasized class 1 recommendations for weight loss, exercise, smoking cessation, minimization or elimination of alcohol consumption, optimal blood pressure control, and a comprehensive care program (**Table 2**) for improvement in outcomes,[4,52-56] which is similar to the recently published European AF guidelines.[51]

Stroke and Cognitive Disease Prevention

In patients with estimated risk of ischemic stroke or thromboembolic events 2% or greater per year (eg, CHA_2DS_2-VASc ≥2 for men, ≥3 for women), benefits of OACs exceed risks of major bleeding.[4] OAC therapy is associated with reduced rates of cognitive impairment and dementia.[57] Several risk scores identify patients at high risk of stroke, with the CHA_2DS_2-VASc score being the most widely validated, which assigns 1 point for congestive HF, 1 point for hypertension, 1 point for age 65 years and older, 2 points for age 75 years and older, 2 points for previous stroke or transient ischemic attack, 1 point for vascular disease, and 1 point for female sex.[4]

In patients without mechanical heart valves or moderate to severe mitral stenosis, DOACs are preferred because they are associated with less bleeding than warfarin. All DOACs reduce risk of intracranial hemorrhage by approximately half compared with warfarin.[41-44,58] After ablation, OACs should be continued for at least 3 months; subsequent decisions should be guided by patients' stroke risk.[4] Ongoing clinical studies are evaluating the safety of discontinuing OACs in patients who have undergone ablation and have no apparent recurrence of AF.[4]

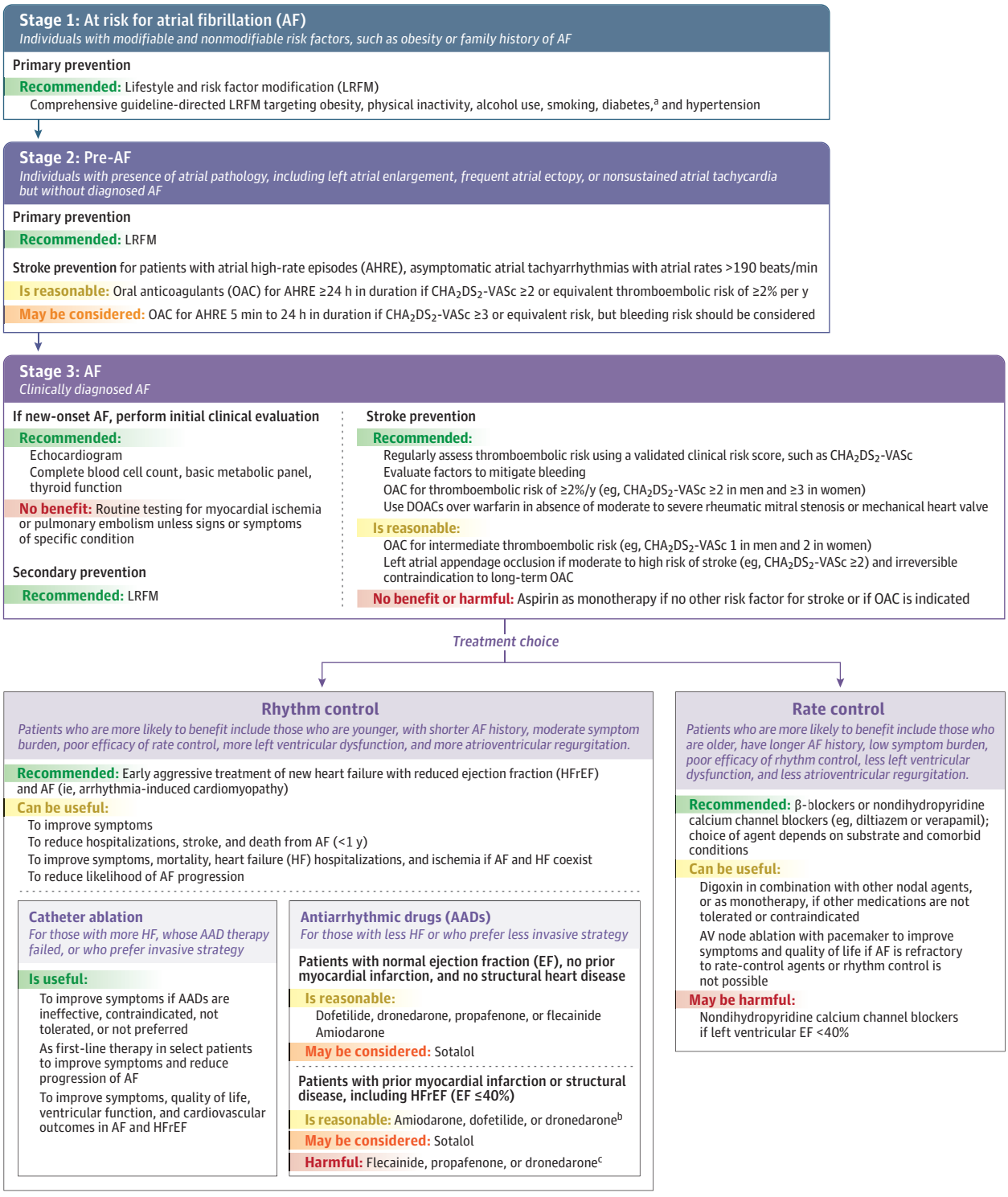
No definitive randomized clinical trials have directly compared the efficacy and safety of 1 DOAC with another.[4] The American Geriatrics Society recommends that because of higher bleeding rates compared with other DOACs, rivaroxaban should be avoided in adults 65 years and older with AF, except when once-daily dosing may improve medication adherence.[59] Apixaban and rivaroxaban are approved for patients receiving dialysis based on limited pharmacokinetic data.[4]

In patients with AF eligible for OACs who do not have a clinical condition requiring treatment with antiplatelet drugs, aspirin alone or aspirin plus clopidogrel as an alternative to OACs are considered harmful because they are less effective than OACs for preventing cardioembolic stroke in AF despite similar bleeding risk.[4,51] Compared with placebo, aspirin does not lower risk of stroke, but is associated with increased major bleeding risk.[4,60] Adding antiplatelet drugs to OACs increases risk of bleeding and is recommended only after acute coronary syndrome or percutaneous coronary intervention (PCI). Based on randomized clinical trials evaluating post-PCI antithrombotic regimens,[61-64]

© 2024 American Medical Association. All rights reserved, including those for text and data mining, AI training, and similar technologies.

Downloaded from jamanetwork.com by Northern Arizona University user on 05/25/2025

Figure 2. Treatment Care Pathway



Treatment care pathway adapted from 2023 ACC/AHA/ACCP/HRS guideline.[4] Color codes represent class of recommendation. Green indicates class 1 benefit>>>risk; yellow, class 2a benefit>>risk; orange, class 2b benefit≥risk; and red, class 3, no benefit or harm. This figure has not been validated for clinical use.

[a]Diabetes management did not receive a class 1 recommendation for secondary prevention in the US AF guideline,[4] but did in the European AF guideline.[51]

[b]Without recent decompensated HF or severe left ventricular dysfunction.

[c]With New York Heart Association class III or IV HF or HF decompensation in past 4 weeks.

© 2024 American Medical Association. All rights reserved, including those for text and data mining, AI training, and similar technologies.

Downloaded from jamanetwork.com by Northern Arizona University user on 05/25/2025

Table 2. Benefits of Adhering to ACC/AHA/ACCP/HRS Guideline Regarding Lifestyle and Risk Factor Modification for Secondary Prevention of Atrial Fibrillation (AF)

			Associated benefits of adhering to recommendations[c]							
Risk factor	Level of evidence[a]	Recommendation[b]	AF symptoms	AF burden[d]	Maintenance sinus rhythm	AF recurrence[e]	AF progression[f]	Functional capacity	Quality of life	AF complications
Overweight or obesity	B-R[4,52]	Weight loss, targeting ≥10% weight reduction	Decreased	Decreased	Increased	Decreased	Decreased			
Physical fitness[g]	B-R[4,53]	Moderate to vigorous exercise training, targeting 210 min per week	Decreased	Decreased	Increased	Decreased		Increased	Improved	Decreased
Cigarette smoking	B-NR[4,54]	Advised to quit smoking and receive goal-directed medical therapy for tobacco cessation								Decreased
Alcohol consumption	B-R[4,55]	Individuals seeking rhythm control should minimize or eliminate alcohol consumption	Decreased	Decreased		Decreased	Decreased			
Hypertension	B-NR[4,56]	Optimal blood pressure control				Decreased				Decreased
Comprehensive care	A[4]	Comprehensive care addressing lifestyle and risk factor modification, AF symptoms, risk of stroke, and other associated medical conditions		Decreased			Decreased			Decreased

Abbreviations: ACC, American College of Cardiology; ACCP, American College of Clinical Pharmacy; AHA, American Heart Association; HRS, Heart Rhythm Society; RCT, randomized clinical trial.

[a] Level of evidence A is high quality from >1 RCT, meta-analysis of high-quality RCTs, or ≥1 RCT corroborated by high-quality registries; B-R, moderate quality from ≥1 RCT or meta-analysis of moderate-quality RCTs; B-NR, moderate-quality evidence from ≥1 well-designed, well-executed nonrandomized study, observational study, or registry study, or meta-analysis of such studies.

[b] Class of recommendation 1 represents the strongest recommendation, as benefit outweighs risk. All recommendations included in the Table are Class 1.

[c] Blank cells indicate that no data were available.

[d] AF burden has several meanings (study dependent), including AF symptom severity and percentage of time spent in AF (duration and number of episodes).

[e] AF recurrence includes studies postablation.

[f] AF progression indicates progression in AF stage (eg, paroxysmal to persistent AF or persistent to permanent).

[g] Exercise training is recommended in patients without AF in the setting of excessive exercise training.

DOACs should be recommended instead of warfarin for patients with AF undergoing PCI.[4] Aspirin should be discontinued early (1-4 weeks) after PCI with continuation of OAC plus P2Y12 inhibitor (clopidogrel preferred over ticagrelor or prasugrel)[65] over triple antithrombotic therapy (OAC, aspirin, and P2Y12 inhibitor) to reduce clinically relevant bleeding risk.[4]

Current guidelines recommend OAC monotherapy over OAC plus single antiplatelet therapy in patients with AF and chronic coronary artery disease who have undergone coronary revascularization more than 1 year previously, unless the patient has a history of prior stent thrombosis.[4] In the Atrial Fibrillation and Ischemic Events With Rivaroxaban in Patients With Stable Coronary Artery Disease (AFIRE) trial, which included 2236 patients with AF, rivaroxaban monotherapy was superior to rivaroxaban plus aspirin or P2Y12 inhibitors for preventing major bleeding (HR, 0.59 [95% CI, 0.39-0.89]; $P = .01$) and noninferior for major cardiovascular events (4.14% and 5.75% per patient-year; HR, 0.72 [95% CI, 0.55-0.95]; $P < .001$).[66]

Although bleeding risk scores, such as HAS-BLED[67] and HEMORR$_2$HAGES,[68] may help inform decision-making regarding initiating OACs in patients with AF and increased risk of bleeding, these scores have limitations because they incorporate factors that are also associated with higher stroke risk. Bleeding risk scores may help identify modifiable risk factors for bleeding and the need for more frequent follow-up.[4,51]

For patients with recurrent bleeding requiring blood transfusions or who have spontaneous intracerebral hemorrhage, OACs may be contraindicated. In these patients, percutaneous left atrial appendage occlusion (pLAAO) with a self-expanding, umbrella-like device is a reasonable alternative to prevent stroke.[4] No direct evidence from randomized clinical trials exists on benefits of pLAAO in OAC-ineligible patients. pLAAO was US Food and Drug Administration–approved based on randomized clinical trials comparing pLAAO with warfarin in warfarin-eligible patients.[69-72] In patients for whom long-term OACs are contraindicated, pLAAO is reasonable with the understanding that it requires at least 45 days of OACs followed by dual antiplatelet therapy for 6 months and lifelong aspirin (325 mg).[4,69-72] Some situations, such as peri-device leak greater than 5 mm or device thrombosis, require prolonged OACs after pLAAO.[69-72]

© 2024 American Medical Association. All rights reserved, including those for text and data mining, AI training, and similar technologies.
Downloaded from jamanetwork.com by Northern Arizona University user on 05/25/2025

Table 3. Randomized Clinical Trials of Catheter Ablation for Atrial Fibrillation (AF)[a]

Source	Enrollment period	Participants, No.	Inclusion criteria	Control	Primary outcome	Length of follow-up, mo
RAAFT-1,[86] 2005	Dec 2001-Jul 2002	Ablation: 33 Control: 37	Symptomatic, untreated paroxysmal AF	AAD	Recurrence of AF >15 s Ablation: 4 (13%) AAD: 22 (63%) (*P* < .001)	12
Oral et al,[87] 2006	Nov 2002-Feb 2004	Ablation + AAD: 77 Control: 69	Persistent AF >6 mo with recurrence within 1 wk of direct current cardioversion	AAD	Freedom from AF/atrial flutter Ablation: 57 (74%) AAD: 40 (58%) (*P* = .05)	12
A4 study,[79] 2008	NA	Ablation: 53 Control: 59	Symptomatic, paroxysmal AF resistant to ≥1 AAD	AAD	Recurrent AF ≥3 min Ablation: 13 (23%) AAD: 46 (89%) (*P* < .001)	12
PABA-CHF,[88] 2008	Nov 2002-Jun 2006	Ablation: 41 Control: 40	Symptomatic paroxysmal or persistent AF resistant to ≥1 AAD, NYHA II or III HF, and LVEF ≤40%	Atrioventricular junction ablation with biventricular pacing	Composite of LVEF, 6MWT, and MLHFQ score Ablation: LVEF 35 ± 9% 6MWT 340 ± 49 m MLHFQ 60 ± 8 Control: 28 ± 6% 6MWT 297 ± 36 m, MLHFQ 82 ± 14 (*P* < .001)	6
ThermoCool AF,[89] 2010	Oct 2004-Oct 2007	Ablation: 106 Control: 61	Symptomatic, paroxysmal AF resistant to ≥1 AAD	AAD	Freedom from protocol-defined treatment failure Ablation: 66% AAD: 16% (HR, 0.30 [95% CI, 0.19-0.47]; *P* < .001)	9
MANTRA-PAF,[90] 2012	Jun 2005-Mar 2009	Ablation: 146 Control: 148	Symptomatic, untreated paroxysmal AF	AAD	Burden of AF (90th percentile) Ablation: 9% AAD: 18% (*P* = .007)	24
STOP AF,[91] 2013	NA	Ablation: 163 Control: 82	Symptomatic, paroxysmal AF resistant to ≥1 AAD	AAD	Freedom from chronic treatment failure Ablation: 114 (69.9%) AAD: 6 (7.3%) (*P* < .001)	12
RAAFT-2,[92] 2014	Jul 2006-Jan 2010	Ablation: 66 Control: 61	Symptomatic, untreated paroxysmal AF	AAD	Time to recurrence of AT >30 s Ablation: 36 (54.5%) AAD: 44 (72.1%) (HR, 0.56 [95% CI, 0.35-0.90]; *P* = .02)	24
AATAC,[93] 2016	NA	Ablation: 102 Control: 101	Persistent AF, NYHA II or III HF, LVEF ≤40%, dual chamber implantable cardioverter defibrillator/cardiac resynchronization therapy with defibrillator	AAD	Freedom from AT >30 s Ablation: 71 (70%) AAD: 34 (34%) (*P* < .001)	24
CASTLE-AF,[94] 2018	Jan 2008-Jan 2016	Ablation: 179 Control: 184	Paroxysmal or persistent AF, resistant to AAD, NYHA ≥II HF, and LVEF ≤35%	AAD or rate control	Composite of death from any cause or HF hospitalization Ablation: 51 (28.5%) AAD/rate control: 82 (44.6%) (HR, 0.62 [95% CI, 0.43-0.87]; *P* = .007)	Median, 37.8
CABANA,[83] 2019	Nov 2009-Apr 2016	Ablation: 1108 Control: 1096	Aged ≥65 y or <65 y with ≥1 stroke risk factor with paroxysmal or persistent AF	AAD or rate control	Death, disabling stroke, serious bleeding, or cardiac arrest Ablation: 89 (8%) Control: 101 (9.2%) (HR, 0.86 [95% CI, 0.65-1.15]; *P* = .30)	Median, 48.5

(continued)

Rate and Rhythm Control

The 2 primary strategies for management of AF are rate control, in which the ventricular rate is slowed with drugs that prolong the AV nodal refractory period, and rhythm control, in which therapeutic interventions aim to restore or maintain sinus rhythm. The choice of rate or rhythm control does not affect the decision for

© 2024 American Medical Association. All rights reserved, including those for text and data mining, AI training, and similar technologies.

Downloaded from jamanetwork.com by Northern Arizona University user on 05/25/2025

Table 3. Randomized Clinical Trials of Catheter Ablation for Atrial Fibrillation (AF)[a] (continued)

Source	Enrollment period	Participants, No.	Inclusion criteria	Control	Primary outcome	Length of follow-up, mo
EAST-AFNET 4,[76] 2020	Jul 2011-Dec 2016	Ablation or AAD: 1395 Control: 1394	AF for ≤12 mo	Rate control	Composite of cardiovascular death, stroke, or hospitalization for HF or acute coronary syndrome; number of nights spent in the hospital per year Rhythm control: 3.9/100 person-years Rate control: 5.0/100 person-years (HR, 0.79 [95% CI, 0.66-0.94]; *P* = .005)	24
EARLY-AF,[80] 2021	Jan 2017-Dec 2018	Ablation: 154 Control: 149	Symptomatic, untreated paroxysmal AF	AAD	Recurrence of AT >30 s Ablation: 66 (42.9%) AAD: 101 (67.8%) (HR, 0.48 [95% CI, 0.35-0.66]; *P* < .001)	12

Abbreviations: 6MWT, 6-Minute Walk Test; AAD, antiarrhythmic drug; AT, atrial tachyarrhythmia (includes atrial fibrillation, atrial flutter, or atrial tachycardia); HF, heart failure; HR, hazard ratio; LVEF, left ventricular ejection fraction; MLHFQ, Minnesota Living with Heart Failure Questionnaire; NA, not available; NYHA, New York Heart Association.

Trial abbreviations: CABANA, Catheter Ablation Versus Antiarrhythmic Drugs for Atrial Fibrillation; CAMTAF, Catheter Ablation Versus Medical Treatment of Atrial Fibrillation in Heart Failure; CASTLE-AF, Catheter Ablation Versus Standard Conventional Treatment in Patients with Left Ventricular Dysfunction and Atrial Fibrillation; EARLY-AF, Early Aggressive Invasive Intervention for Atrial Fibrillation; EAST-AFNET 4, Early Treatment of Atrial Fibrillation for Stroke Prevention Trial; MANTRA-PAF, Radiofrequency Ablation as Initial Therapy in Paroxysmal Atrial Fibrillation; PABA-CHF, Pulmonary Vein Antrum Isolation vs AV Node Ablation with Biventricular Pacing for Treatment of Atrial Fibrillation in Patients with Congestive Heart Failure; RAAFT, Radiofrequency Ablation vs Antiarrhythmic Drugs as First-Line Treatment of Symptomatic Atrial Fibrillation; ThermoCool AF, Comparison of Antiarrhythmic Drug Therapy and Radiofrequency Catheter Ablation in Patients With Paroxysmal Atrial Fibrillation.

[a] Randomized clinical trials were selected based on citation frequency and their impact on shaping clinical guidelines.

or duration of OAC. Rhythm control interventions consist of AADs, cardioversion, and/or ablation. The Atrial Fibrillation Follow-up Investigation of Rhythm Management (AFFIRM) and Rate Control vs Electrical Cardioversion for Atrial Fibrillation (RACE) studies compared AF strategies and showed no significant differences in clinical outcomes, including mortality and stroke.[73,74]

More recent studies reported the benefits of initiating rhythm control within 1 year of AF diagnosis (early rhythm control) over rate control for reducing HF, stroke risk, and mortality in patients with paroxysmal and persistent AF.[75-77] The benefit of early rhythm control was also observed in those with asymptomatic AF.

Rate control without attempts to treat rhythm is an appropriate strategy for patients who are unlikely to benefit from rhythm control with AADs or catheter ablation and for those in whom rhythm control is considered too risky. For example, patients with amyloid cardiomyopathy or cor pulmonale are more likely to have recurrent AF following ablation, therefore rate control may be more appropriate. Rate control is also appropriate for patients with stage 4 (permanent) AF. Rate control is achieved with single or drug combinations, including β-blockers such as metoprolol, esmolol, or atenolol, and nondihydropyridine calcium channel blockers, such as verapamil or diltiazem, to slow electrical conduction through the atrioventricular node.[4] Digoxin can be used as adjunctive therapy when ventricular rate remains poorly controlled or hypotension limits further titration of β-blockers or nondihydropyridine calcium channel blockers.[4] Drugs that slow atrioventricular conduction should be titrated to control symptoms and achieve resting heart rates less than 100 to 110 beats per minute.[4] Atrioventricular nodal ablation with pacemaker implant can be useful for controlling symptoms and improving quality of life (QOL) in patients with ineffectively controlled ventricular rates who have not improved or are not candidates for rhythm control.[4]

Pharmacologic Rhythm Control

AADs may be used for acute conversion of AF to sinus rhythm ("pill-in-the-pocket") or for suppression of AF when taken daily. Pill-in-the-pocket treatment with flecainide or propafenone in conjunction with concomitant AV nodal agent(s) may be useful for intermittent treatment of infrequent AF episodes, but efficacy for acute conversion and safety (proarrhythmic effect) must first be tested in a monitored hospital setting.[51] Patients should be observed for at least 8 hours after dose.[4]

Early rate vs rhythm randomized clinical trials did not demonstrate improved mortality or stroke risk with daily use of AAD compared with treatment with AV nodal agents.[78] However, the 2020 Early Treatment of Atrial Fibrillation for Stroke Prevention Trial (EAST-AFNET 4) reported clinical benefit of rhythm control with daily AAD or catheter ablation over rate control therapy when rhythm control was initiated early (ie, within 12 months of AF diagnosis).[76] EAST-AFNET 4 randomized 2789 patients within 1 year of AF onset (median time, 36 days) and mean CHA_2DS_2-VASc of 3.4 to rhythm control or rate control. Most patients (87%) were initially treated with AADs, including flecainide (35.9%), amiodarone (19.6%), and dronedarone (16.7%), whereas 19.4% underwent ablation by 2 years. The trial was stopped early due to a 21% reduction in the primary composite outcome of cardiovascular mortality, stroke, and hospitalizations for HF or acute coronary syndrome with rhythm control vs rate control (3.9 per 100

© 2024 American Medical Association. All rights reserved, including those for text and data mining, AI training, and similar technologies.
Downloaded from jamanetwork.com by Northern Arizona University user on 05/25/2025

person-years vs 5.0 per 100 person-years).[76] The EAST-AFNET 4 trial reported that benefit of rhythm control extended to those with asymptomatic AF.[77] AADs are not atrial cardiomyocyte selective and can alter ventricular myocardial electrophysiologic properties; therefore, the presence of comorbidities, such as prior myocardial infarction or HFrEF (40%), influences AAD selection (Figure 2).

Catheter Ablation

The discovery that focal ectopic impulses arising from the pulmonary veins often initiate AF and that ablation of these sources significantly reduced AF preceded development of catheter ablation procedures, which are effective for maintaining sinus rhythm (Table 1). Ablation destroys atrial tissue at the atrial–pulmonary vein junction to prevent ectopic impulses from the pulmonary vein from reaching the atrium (ie, pulmonary vein isolation). The A4 study randomized 112 patients with paroxysmal AF to treatment with AAD or radiofrequency ablation.[79] The primary end point was longer than 3 minutes of AF or reported AF symptoms. At 1-year follow-up, 89% of patients who underwent ablation remained AF-free vs 23% of patients treated with AAD ($P < .0001$). A significant improvement in symptoms, QOL, and exercise capacity was also observed with ablation.[79]

The Early Aggressive Invasive Intervention for Atrial Fibrillation (EARLY-AF) trial randomized 303 patients with symptomatic, untreated paroxysmal AF to cryoablation or AAD and monitored for recurrent AF using ILR.[80] The primary end point was first recurrence of AF (or atrial tachyarrhythmia) of 30 seconds or longer; the secondary end point was overall burden of AF. The decision for OAC was based on risk of stroke (CHA_2DS_2-VASc ≥1) and irrespective of group assignment. At 1-year follow-up, recurrence of AF was significantly less likely in those who underwent ablation (42.9%) compared with AADs (67.8%; HR, 0.48 [95% CI, 0.35-0.66]). The mean time in AF was significantly lower with ablation (0.6%) vs AADs (3.9%). Meta-analysis of 5 clinical trials that enrolled 693 people with AF (predominately paroxysmal AF), reported a more than 2-fold greater freedom from AF at 1 year with radiofrequency ablation (77%) vs AAD (29%).[81] Ablation in patients with persistent AF (including long-standing persistent) with low prevalence of structural heart disease (eg, normal left ventricular function, normal to mild left atrial enlargement, and low prevalence of coronary artery disease or diabetes) has comparable results to those with paroxysmal AF.[82]

The Catheter Ablation Versus Antiarrhythmic Drugs for Atrial Fibrillation (CABANA) trial tested efficacy of ablation on clinical outcomes. It enrolled 2204 patients with paroxysmal or persistent AF and median CHA_2DS_2-VASc score of 3 who had previously been treated with 2 or more AADs. All patients irrespective of treatment group received anticoagulation based on stroke risk (CHA_2DS_2-VASc ≥2). CABANA reported that ablation did not significantly reduce the primary composite end point of death, disabling stroke, serious bleeding, or cardiac arrest.[83] The statistical power of the study was limited by lower-than-expected event rates, high crossover rates, and loss to follow-up. Despite this, CABANA reported significant improvement in QOL with ablation.[84] In prespecified subgroup analysis of CABANA, compared with medical treatment, patients who received ablation (intention-to-treat analysis) younger than 65 years had lower mortality, whereas those 75 years or older did not.[85]

Ablation in HF

Coexistence of HF and AF is associated with increased mortality compared with those with HF or AF alone.[37] Although AADs have limited benefit in AF patients with HFrEF, multiple randomized clinical trials and meta-analyses have shown superiority of rhythm control with ablation over medical therapy (**Table 3**).[88,93-100] The Catheter Ablation Versus Standard Conventional Treatment in Patients with Left Ventricular Dysfunction and Atrial Fibrillation (CASTLE-AF) trial randomized 363 patients with HF and left ventricular ejection fraction 35% or less to ablation or medical therapy.[94] After median follow-up of 38 months, ablation was associated with significant reductions in the composite end point of death or hospitalization for HF (ablation, 28.5% vs medical therapy, 44.6%; HR, 0.62 [95% CI, 0.43-0.87]). A meta-analysis of 3 trials that included 977 patients with HF and AF reported that compared with medical therapy, ablation was associated with lower mortality rates (RR, 0.61 [95% CI, 0.44-0.84]) and HF hospitalizations (RR, 0.60 [95% CI, 0.49-0.74]; absolute rates unavailable).[101]

Few randomized clinical trials have studied the effects of ablation in people with AF and HFpEF. CABANA was the only large randomized clinical trial (n = 2204) in which nearly 80% of patients with HF had an ejection fraction of 50% or greater.[102] In a prespecified subgroup analysis of patients with New York Heart Association class greater than 2, ablation was associated with 43% reduction in all-cause mortality (6.1% ablation vs 9.3% medical therapy), 44% reduction in AF recurrence (56% ablation vs 72% medical therapy), and sustained improvement of QOL out to 5 years (adjusted mean difference, 5 points).[102]

Inequities in AF Management and Outcomes

AF management and outcomes are associated with inequities by sex, race and ethnicity, and social determinants of health (SDOH), defined as nonmedical factors that influence health. Individuals with AF who are women, Black, or Hispanic; with lower income or education; inadequate or lack of insurance; or who live in rural areas or neighborhoods characterized by material deprivation are less likely to receive guideline-directed care and more likely to experience worse outcomes.[4,103] A study from Ontario, Canada, reported that 1 year after AF diagnosis, patients residing in neighborhoods with the highest compared with the lowest material deprivation were less likely to have cardiology visits (27.9% in quintile 5 vs 34% in quintile 1; aHR, 0.84), were less likely to receive guideline-based therapies, including OACs (53.9% quintile 5 vs 56.8% in quintile 1; aHR, 0.97) and ablation (0.1% in quintile 5 vs 0.3% in quintile 1; aHR, 0.45), and experienced worse outcomes, including more strokes (1.8% in quintile 5 vs 1.4% in quintile 1; aHR, 1.15) and higher mortality (17.9% in quintile 5 vs 14.1% in quintile 1; aHR, 1.16).[104]

The US Centers for Medicare & Medicaid Services and the Joint Commission have instituted SDOH reporting requirements with the goal of reducing health inequities.[105,106]

Limitations

This review has limitations. First, relevant studies may not have been included. Second, generalizability of the evidence is limited because many clinical trials have not included substantial proportions of participants who were Black or Hispanic, and have not examined variation by SDOH.[107] Third, the quality of included evidence was not systematically reviewed.

© 2024 American Medical Association. All rights reserved, including those for text and data mining, AI training, and similar technologies.

Downloaded from jamanetwork.com by Northern Arizona University user on 05/25/2025

Conclusions

AF is associated with increased rates of stroke, heart failure, and mortality. Guideline-directed AF management includes lifestyle and risk factor modification to prevent AF onset, recurrence, and complications, and OACs are recommended for those with an estimated risk of stroke or thromboembolic events of 2% or greater per year. Early rhythm control using AADs or catheter ablation is recommended in select patients with AF who have symptomatic paroxysmal AF or HFrEF.

ARTICLE INFORMATION

Accepted for Publication: October 8, 2024.

Published Online: December 16, 2024. doi:10.1001/jama.2024.22451

Author Affiliations: Hinda and Arthur Marcus Institute for Aging Research, Hebrew SeniorLife, Harvard Medical School, Boston, Massachusetts (Ko); Section of Cardiovascular Medicine, Department of Medicine, Boston Medical Center, Boston University Chobanian & Avedisian School of Medicine, Boston, Massachusetts (Ko, Evans, Benjamin, Helm); Smith Center for Outcomes Research in Cardiology, Beth Israel Deaconess Medical Center, Harvard Medical School, Boston, Massachusetts (Ko); Department of Cardiovascular Medicine, Heart, Vascular & Thoracic Institute, Cleveland Clinic, Cleveland, Ohio (Chung); Department of Cardiovascular and Metabolic Sciences, Lerner Research Institute, Cleveland Clinic, Cleveland, Ohio (Chung); Cleveland Clinic Lerner College of Medicine of Case Western Reserve University, Cleveland, Ohio (Chung); Department of Epidemiology, Boston University School of Public Health, Boston, Massachusetts (Benjamin).

Conflict of Interest Disclosures: Dr Ko reported receiving grants to institution from Boston Scientific; grants from National Institutes of Health (NIH); and personal fees from Windrose Consulting Group and AcademicCME outside the submitted work. Dr Chung reported receiving research grants from NIH and American Heart Association (AHA) during the conduct of the study. Dr Benjamin reported receiving grants from NIH/National Heart, Lung, and Blood Institute (R01HL092577) and AHA (AHA_18SFRN34110082) during the conduct of the study. Dr Helm reported receiving grants to institution from Boston Scientific and Cardathea outside the submitted work. No other disclosures were reported.

Submissions: We encourage authors to submit papers for consideration as a Review. Please contact Kristin Walter, MD, at kristin.walter@jamanetwork.org.

REFERENCES

1. Mou L, Norby FL, Chen LY, et al. Lifetime risk of atrial fibrillation by race and socioeconomic status: ARIC study (Atherosclerosis Risk in Communities). *Circ Arrhythm Electrophysiol*. 2018;11(7):e006350. doi:10.1161/CIRCEP.118.006350

2. Weng LC, Preis SR, Hulme OL, et al. Genetic predisposition, clinical risk factor burden, and lifetime risk of atrial fibrillation. *Circulation*. 2018;137(10):1027-1038. doi:10.1161/CIRCULATIONAHA.117.031431

3. Noubiap JJ, Tang JJ, Teraoka JT, Dewland TA, Marcus GM. Minimum national prevalence of diagnosed atrial fibrillation inferred from California acute care facilities. *J Am Coll Cardiol*. 2024;84(16):1501-1508. doi:10.1016/j.jacc.2024.07.014

4. Joglar JA, Chung MK, Armbruster AL, et al; Peer Review Committee Members. 2023 ACC/AHA/ACCP/HRS guideline for the diagnosis and management of atrial fibrillation: a report of the American College of Cardiology/American Heart Association Joint Committee on Clinical Practice Guidelines. *Circulation*. 2024;149(1):e1-e156. doi:10.1161/CIR.0000000000001193

5. Martin SS, Aday AW, Almarzooq ZI, et al; American Heart Association Council on Epidemiology and Prevention Statistics Committee and Stroke Statistics Subcommittee. 2024 heart disease and stroke statistics: a report of US and global data from the American Heart Association. *Circulation*. 2024;149(8):e347-e913. doi:10.1161/CIR.0000000000001209

6. Vinter N, Cordsen P, Johnsen SP, et al. Temporal trends in lifetime risks of atrial fibrillation and its complications between 2000 and 2022: Danish, nationwide, population based cohort study. *BMJ*. 2024;385:e077209. doi:10.1136/bmj-2023-077209

7. Schnabel RB, Yin X, Gona P, et al. 50 year trends in atrial fibrillation prevalence, incidence, risk factors, and mortality in the Framingham Heart Study: a cohort study. *Lancet*. 2015;386(9989):154-162. doi:10.1016/S0140-6736(14)61774-8

8. Alonso A, Krijthe BP, Aspelund T, et al. Simple risk model predicts incidence of atrial fibrillation in a racially and geographically diverse population: the CHARGE-AF consortium. *J Am Heart Assoc*. 2013;2(2):e000102. doi:10.1161/JAHA.112.000102

9. Himmelreich JCL, Veelers L, Lucassen WAM, et al. Prediction models for atrial fibrillation applicable in the community: a systematic review and meta-analysis. *Europace*. 2020;22(5):684-694. doi:10.1093/europace/euaa005

10. Khurshid S, Kartoun U, Ashburner JM, et al. Performance of atrial fibrillation risk prediction models in over 4 million individuals. *Circ Arrhythm Electrophysiol*. 2021;14(1):e008997. doi:10.1161/CIRCEP.120.008997

11. Zhang D, Ma Y, Xu J, Yi F. Association between obstructive sleep apnea (OSA) and atrial fibrillation (AF): a dose-response meta-analysis. *Medicine (Baltimore)*. 2022;101(30):e29443. doi:10.1097/MD.0000000000029443

12. Huang M, Yang S, Ge G, Zhi H, Wang L. Effects of thyroid dysfunction and the thyroid-stimulating hormone levels on the risk of atrial fibrillation: a systematic review and dose-response meta-analysis from cohort studies. *Endocr Pract*. 2022;28(8):822-831. doi:10.1016/j.eprac.2022.05.008

13. Jiang H, Mei X, Jiang Y, et al. Alcohol consumption and atrial fibrillation risk: an updated dose-response meta-analysis of over 10 million participants. *Front Cardiovasc Med*. 2022;9:979982. doi:10.3389/fcvm.2022.979982

14. Valenzuela PL, Morales JS, Santos-Lozano A, et al. What do we really know about the association between physical activity, sports, and atrial fibrillation? A systematic review and meta-analysis from unbiased studies. *Eur J Prev Cardiol*. 2022;29(4):e143-e148. doi:10.1093/eurjpc/zwab073

15. Wang Q, Richardson TG, Sanderson E, et al. A phenome-wide bidirectional Mendelian randomization analysis of atrial fibrillation. *Int J Epidemiol*. 2022;51(4):1153-1166. doi:10.1093/ije/dyac041

16. McCauley MD, Iacobellis G, Li N, Nattel S, Goldberger JJ. Targeting the substrate for atrial fibrillation: JACC review topic of the week. *J Am Coll Cardiol*. 2024;83(20):2015-2027. doi:10.1016/j.jacc.2024.02.050

17. Staerk L, Sherer JA, Ko D, Benjamin EJ, Helm RH. Atrial fibrillation: epidemiology, pathophysiology, and clinical outcomes. *Circ Res*. 2017;120(9):1501-1517. doi:10.1161/CIRCRESAHA.117.309732

18. Goette A, Kalman JM, Aguinaga L, et al; Document Reviewers. EHRA/HRS/APHRS/SOLAECE expert consensus on atrial cardiomyopathies: definition, characterization, and clinical implication. *Europace*. 2016;18(10):1455-1490. doi:10.1093/europace/euw161

19. Svendsen JH, Diederichsen SZ, Højberg S, et al. Implantable loop recorder detection of atrial fibrillation to prevent stroke (The LOOP Study): a randomised controlled trial. *Lancet*. 2021;398(10310):1507-1516. doi:10.1016/S0140-6736(21)01698-6

20. Perez MV, Mahaffey KW, Hedlin H, et al; Apple Heart Study Investigators. Large-scale assessment of a smartwatch to identify atrial fibrillation. *N Engl J Med*. 2019;381(20):1909-1917. doi:10.1056/NEJMoa1901183

21. Hart RG, Sharma M, Mundl H, et al; NAVIGATE ESUS Investigators. Rivaroxaban for stroke prevention after embolic stroke of undetermined source. *N Engl J Med*. 2018;378(23):2191-2201. doi:10.1056/NEJMoa1802686

22. Diener HC, Sacco RL, Easton JD, et al; RE-SPECT ESUS Steering Committee and Investigators. Dabigatran for prevention of stroke after embolic stroke of undetermined source. *N Engl J Med*. 2019;380(20):1906-1917. doi:10.1056/NEJMoa1813959

23. Sanna T, Diener HC, Passman RS, et al; CRYSTAL AF Investigators. Cryptogenic stroke and underlying atrial fibrillation. *N Engl J Med*. 2014;370(26):2478-2486. doi:10.1056/NEJMoa1313600

24. Bernstein RA, Kamel H, Granger CB, et al; STROKE-AF Investigators. Effect of long-term continuous cardiac monitoring vs usual care on detection of atrial fibrillation in patients with stroke attributed to large- or small-vessel disease: the STROKE-AF randomized clinical trial. *JAMA*. 2021;325(21):2169-2177. doi:10.1001/jama.2021.6470

25. Buck BH, Hill MD, Quinn FR, et al. Effect of implantable vs prolonged external electrocardiographic monitoring on atrial fibrillation detection in patients with ischemic stroke: the PER DIEM randomized clinical trial. *JAMA*. 2021;325(21):2160-2168. doi:10.1001/jama.2021.6128

© 2024 American Medical Association. All rights reserved, including those for text and data mining, AI training, and similar technologies.

Downloaded from jamanetwork.com by Northern Arizona University user on 05/25/2025

26. Chyou JY, Barkoudah E, Dukes JW, et al; American Heart Association Acute Cardiac Care and General Cardiology Committee, Electrocardiography and Arrhythmias Committee, and Clinical Pharmacology Committee of the Council on Clinical Cardiology; Council on Cardiovascular Surgery and Anesthesia; Council on Cardiopulmonary, Critical Care, Perioperative and Resuscitation; Council on Cardiovascular and Stroke Nursing; and Stroke Council. Atrial fibrillation occurring during acute hospitalization: a scientific statement from the American Heart Association. *Circulation*. 2023;147(15):e676-e698. doi:10.1161/CIR.0000000000001133

27. Walkey AJ, Wiener RS, Ghobrial JM, Curtis LH, Benjamin EJ. Incident stroke and mortality associated with new-onset atrial fibrillation in patients hospitalized with severe sepsis. *JAMA*. 2011;306(20):2248-2254. doi:10.1001/jama.2011.1615

28. Lubitz SA, Yin X, Rienstra M, et al. Long-term outcomes of secondary atrial fibrillation in the community: the Framingham Heart Study. *Circulation*. 2015;131(19):1648-1655. doi:10.1161/CIRCULATIONAHA.114.014058

29. McIntyre WF, Um KJ, Cheung CC, et al. Atrial fibrillation detected initially during acute medical illness: a systematic review. *Eur Heart J Acute Cardiovasc Care*. 2019;8(2):130-141. doi:10.1177/2048872618799748

30. Li S, Lip GYH, Ko D, Walkey A. Anticoagulation for secondary atrial fibrillation. *N Engl J Med*. 2024;390(13):1237-1240. doi:10.1056/NEJMclde2311699

31. Healey JS, Connolly SJ, Gold MR, et al; ASSERT Investigators. Subclinical atrial fibrillation and the risk of stroke. *N Engl J Med*. 2012;366(2):120-129. doi:10.1056/NEJMoa1105575

32. Healey JS, Alings M, Ha A, et al; ASSERT-II Investigators. Subclinical atrial fibrillation in older patients. *Circulation*. 2017;136(14):1276-1283. doi:10.1161/CIRCULATIONAHA.117.028845

33. Mahajan R, Perera T, Elliott AD, et al. Subclinical device-detected atrial fibrillation and stroke risk: a systematic review and meta-analysis. *Eur Heart J*. 2018;39(16):1407-1415. doi:10.1093/eurheartj/ehx731

34. McIntyre WF, Benz AP, Becher N, et al. Direct oral anticoagulants for stroke prevention in patients with device-detected atrial fibrillation: a study-level meta-analysis of the NOAH-AFNET 6 and ARTESiA trials. *Circulation*. 2023;149(13):981-988. doi:10.1161/CIRCULATIONAHA.123.067512

35. Siontis KC, Gersh BJ, Killian JM, et al. Typical, atypical, and asymptomatic presentations of new-onset atrial fibrillation in the community: characteristics and prognostic implications. *Heart Rhythm*. 2016;13(7):1418-1424. doi:10.1016/j.hrthm.2016.03.003

36. Flaker GC, Belew K, Beckman K, et al; AFFIRM Investigators. Asymptomatic atrial fibrillation: demographic features and prognostic information from the Atrial Fibrillation Follow-up Investigation of Rhythm Management (AFFIRM) study. *Am Heart J*. 2005;149(4):657-663. doi:10.1016/j.ahj.2004.06.032

37. Wang TJ, Larson MG, Levy D, et al. Temporal relations of atrial fibrillation and congestive heart failure and their joint influence on mortality: the Framingham Heart Study. *Circulation*. 2003;107(23):2920-2925. doi:10.1161/01.CIR.0000072767.89944.6E

38. Odutayo A, Wong CX, Hsiao AJ, Hopewell S, Altman DG, Emdin CA. Atrial fibrillation and risks of cardiovascular disease, renal disease, and death: systematic review and meta-analysis. *BMJ*. 2016;354:i4482. doi:10.1136/bmj.i4482

39. Piccini JP, Hammill BG, Sinner MF, et al. Clinical course of atrial fibrillation in older adults: the importance of cardiovascular events beyond stroke. *Eur Heart J*. 2014;35(4):250-256. doi:10.1093/eurheartj/eht483

40. Papanastasiou CA, Theochari CA, Zareifopoulos N, et al. Atrial fibrillation is associated with cognitive impairment, all-cause dementia, vascular dementia, and Alzheimer's disease: a systematic review and meta-analysis. *J Gen Intern Med*. 2021;36(10):3122-3135. doi:10.1007/s11606-021-06954-8

41. Connolly SJ, Ezekowitz MD, Yusuf S, et al; RE-LY Steering Committee and Investigators. Dabigatran versus warfarin in patients with atrial fibrillation. *N Engl J Med*. 2009;361(12):1139-1151. doi:10.1056/NEJMoa0905561

42. Granger CB, Alexander JH, McMurray JJV, et al; ARISTOTLE Committees and Investigators. Apixaban versus warfarin in patients with atrial fibrillation. *N Engl J Med*. 2011;365(11):981-992. doi:10.1056/NEJMoa1107039

43. Patel MR, Mahaffey KW, Garg J, et al; ROCKET AF Investigators. Rivaroxaban versus warfarin in nonvalvular atrial fibrillation. *N Engl J Med*. 2011;365(10):883-891. doi:10.1056/NEJMoa1009638

44. Giugliano RP, Ruff CT, Braunwald E, et al; ENGAGE AF-TIMI 48 Investigators. Edoxaban versus warfarin in patients with atrial fibrillation. *N Engl J Med*. 2013;369(22):2093-2104. doi:10.1056/NEJMoa1310907

45. Chamberlain AM, Gersh BJ, Alonso A, et al. No decline in the risk of heart failure after incident atrial fibrillation: a community study assessing trends overall and by ejection fraction. *Heart Rhythm*. 2017;14(6):791-798. doi:10.1016/j.hrthm.2017.01.031

46. Soliman EZ, Rahman AF, Zhang ZM, et al. Effect of intensive blood pressure lowering on the risk of atrial fibrillation. *Hypertension*. 2020;75(6):1491-1496. doi:10.1161/HYPERTENSIONAHA.120.14766

47. Zheng RJ, Wang Y, Tang JN, Duan JY, Yuan MY, Zhang JY. Association of SGLT2 inhibitors with risk of atrial fibrillation and stroke in patients with and without type 2 diabetes: a systemic review and meta-analysis of randomized controlled trials. *J Cardiovasc Pharmacol*. 2022;79(2):e145-e152. doi:10.1097/FJC.0000000000001183

48. January CT, Wann LS, Calkins H, et al. 2019 AHA/ACC/HRS focused update of the 2014 AHA/ACC/HRS guideline for the management of patients with atrial fibrillation: a report of the American College of Cardiology/American Heart Association Task Force on Clinical Practice Guidelines and the Heart Rhythm Society in collaboration with the Society of Thoracic Surgeons. *Circulation*. 2019;140(2):e125-e151. doi:10.1161/CIR.0000000000000665

49. Loring Z, Shrader P, Allen LA, et al. Guideline-directed therapies for comorbidities and clinical outcomes among individuals with atrial fibrillation. *Am Heart J*. 2020;219:21-30. doi:10.1016/j.ahj.2019.10.008

50. Sardana M, Tang Y, Magnani JW, et al. Provider-level variation in smoking cessation assistance provided in the cardiology clinics: insights from the NCDR PINNACLE Registry. *J Am Heart Assoc*. 2019;8(13):e011412. doi:10.1161/JAHA.118.011307

51. Van Gelder IC, Rienstra M, Bunting KV, et al; ESC Scientific Document Group. 2024 ESC guidelines for the management of atrial fibrillation developed in collaboration with the European Association for Cardio-Thoracic Surgery (EACTS). *Eur Heart J*. 2024;45(36):3314-3414. doi:10.1093/eurheartj/ehae176

52. Abed HS, Wittert GA, Leong DP, et al. Effect of weight reduction and cardiometabolic risk factor management on symptom burden and severity in patients with atrial fibrillation: a randomized clinical trial. *JAMA*. 2013;310(19):2050-2060. doi:10.1001/jama.2013.280521

53. Elliott AD, Verdicchio CV, Mahajan R, et al. An exercise and physical activity program in patients with atrial fibrillation: the ACTIVE-AF randomized controlled trial. *JACC Clin Electrophysiol*. 2023;9(4):455-465. doi:10.1016/j.jacep.2022.12.002

54. Lee SR, Choi EK, Jung JH, Han KD, Oh S, Lip GYH. Smoking cessation after diagnosis of new-onset atrial fibrillation and the risk of stroke and death. *J Clin Med*. 2021;10(11):2238. doi:10.3390/jcm10112238

55. Voskoboinik A, Kalman JM, De Silva A, et al. Alcohol abstinence in drinkers with atrial fibrillation. *N Engl J Med*. 2020;382(1):20-28. doi:10.1056/NEJMoa1817591

56. Pinho-Gomes AC, Azevedo L, Copland E, et al; Blood Pressure Lowering Treatment Trialists' Collaboration. Blood pressure-lowering treatment for the prevention of cardiovascular events in patients with atrial fibrillation: an individual participant data meta-analysis. *PLoS Med*. 2021;18(6):e1003599. doi:10.1371/journal.pmed.1003599

57. Cheng W, Liu W, Li B, Li D. Relationship of anticoagulant therapy with cognitive impairment among patients with atrial fibrillation: a meta-analysis and systematic review. *J Cardiovasc Pharmacol*. 2018;71(6):380-387. doi:10.1097/FJC.0000000000000575

58. Ruff CT, Giugliano RP, Braunwald E, et al. Comparison of the efficacy and safety of new oral anticoagulants with warfarin in patients with atrial fibrillation: a meta-analysis of randomised trials. *Lancet*. 2014;383(9921):955-962. doi:10.1016/S0140-6736(13)62343-0

59. 2023 American Geriatrics Society Beers Criteria® Update Expert Panel. American Geriatrics Society 2023 updated AGS Beers Criteria® for potentially inappropriate medication use in older adults. *J Am Geriatr Soc*. 2023;71(7):2052-2081. doi:10.1111/jgs.18372

60. Sato H, Ishikawa K, Kitabatake A, et al; Japan Atrial Fibrillation Stroke Trial Group. Low-dose aspirin for prevention of stroke in low-risk patients with atrial fibrillation: Japan Atrial Fibrillation Stroke Trial. *Stroke*. 2006;37(2):447-451. doi:10.1161/01.STR.0000198839.61112.ee

61. Dewilde WJ, Oirbans T, Verheugt FW, et al; WOEST study investigators. Use of clopidogrel with or without aspirin in patients taking oral anticoagulant therapy and undergoing percutaneous coronary intervention: an open-label, randomised, controlled trial. *Lancet*. 2013;381(9872):1107-1115. doi:10.1016/S0140-6736(12)62177-1

© 2024 American Medical Association. All rights reserved, including those for text and data mining, AI training, and similar technologies.

Downloaded from jamanetwork.com by Northern Arizona University user on 05/25/2025

62. Cannon CP, Bhatt DL, Oldgren J, et al; RE-DUAL PCI Steering Committee and Investigators. Dual antithrombotic therapy with dabigatran after PCI in atrial fibrillation. *N Engl J Med*. 2017;377(16):1513-1524. doi:10.1056/NEJMoa1708454

63. Gibson CM, Mehran R, Bode C, et al. Prevention of bleeding in patients with atrial fibrillation undergoing PCI. *N Engl J Med*. 2016;375(25):2423-2434. doi:10.1056/NEJMoa1611594

64. Lopes RD, Heizer G, Aronson R, et al; AUGUSTUS Investigators. Antithrombotic therapy after acute coronary syndrome or PCI in atrial fibrillation. *N Engl J Med*. 2019;380(16):1509-1524. doi:10.1056/NEJMoa1817083

65. Kumbhani DJ, Cannon CP, Beavers CJ, et al. 2020 ACC expert consensus decision pathway for anticoagulant and antiplatelet therapy in patients with atrial fibrillation or venous thromboembolism undergoing percutaneous coronary intervention or with atherosclerotic cardiovascular disease: a report of the American College of Cardiology Solution Set Oversight Committee. *J Am Coll Cardiol*. 2021;77(5):629-658. doi:10.1016/j.jacc.2020.09.011

66. Yasuda S, Kaikita K, Akao M, et al; AFIRE Investigators. Antithrombotic therapy for atrial fibrillation with stable coronary disease. *N Engl J Med*. 2019;381(12):1103-1113. doi:10.1056/NEJMoa1904143

67. Lip GY, Frison L, Halperin JL, Lane DA. Comparative validation of a novel risk score for predicting bleeding risk in anticoagulated patients with atrial fibrillation: the HAS-BLED (Hypertension, Abnormal Renal/Liver Function, Stroke, Bleeding History or Predisposition, Labile INR, Elderly, Drugs/Alcohol Concomitantly) score. *J Am Coll Cardiol*. 2011;57(2):173-180. doi:10.1016/j.jacc.2010.09.024

68. Apostolakis S, Lane DA, Guo Y, Buller H, Lip GY. Performance of the HEMORR(2)HAGES, ATRIA, and HAS-BLED bleeding risk-prediction scores in patients with atrial fibrillation undergoing anticoagulation: the AMADEUS (evaluating the use of SR34006 compared to warfarin or acenocoumarol in patients with atrial fibrillation) study. *J Am Coll Cardiol*. 2012;60(9):861-867. doi:10.1016/j.jacc.2012.06.019

69. Holmes DR, Reddy VY, Turi ZG, et al; PROTECT AF Investigators. Percutaneous closure of the left atrial appendage versus warfarin therapy for prevention of stroke in patients with atrial fibrillation: a randomised non-inferiority trial. *Lancet*. 2009;374(9689):534-542. doi:10.1016/S0140-6736(09)61343-X

70. Reddy VY, Sievert H, Halperin J, et al; PROTECT AF Steering Committee and Investigators. Percutaneous left atrial appendage closure vs warfarin for atrial fibrillation: a randomized clinical trial. *JAMA*. 2014;312(19):1988-1998. doi:10.1001/jama.2014.15192

71. Holmes DR Jr, Kar S, Price MJ, et al. Prospective randomized evaluation of the Watchman left atrial appendage closure device in patients with atrial fibrillation versus long-term warfarin therapy: the PREVAIL trial. *J Am Coll Cardiol*. 2014;64(1):1-12. doi:10.1016/j.jacc.2014.04.029

72. US Food and Drug Administration. Watchman® LAA closure technology summary of safety and effectiveness data (SSED). March 13, 2015. Accessed March 17, 2024. https://www.accessdata.fda.gov/cdrh_docs/pdf13/P130013B.pdf

73. Wyse DG, Waldo AL, DiMarco JP, et al; Atrial Fibrillation Follow-up Investigation of Rhythm Management (AFFIRM) Investigators. A comparison of rate control and rhythm control in patients with atrial fibrillation. *N Engl J Med*. 2002;347(23):1825-1833. doi:10.1056/NEJMoa021328

74. Van Gelder IC, Hagens VE, Bosker HA, et al; Rate Control versus Electrical Cardioversion for Persistent Atrial Fibrillation Study Group. A comparison of rate control and rhythm control in patients with recurrent persistent atrial fibrillation. *N Engl J Med*. 2002;347(23):1834-1840. doi:10.1056/NEJMoa021375

75. Bunch TJ, May HT, Bair TL, et al. Increasing time between first diagnosis of atrial fibrillation and catheter ablation adversely affects long-term outcomes. *Heart Rhythm*. 2013;10(9):1257-1262. doi:10.1016/j.hrthm.2013.05.013

76. Kirchhof P, Camm AJ, Goette A, et al; EAST-AFNET 4 Trial Investigators. Early rhythm-control therapy in patients with atrial fibrillation. *N Engl J Med*. 2020;383(14):1305-1316. doi:10.1056/NEJMoa2019422

77. Willems S, Borof K, Brandes A, et al. Systematic, early rhythm control strategy for atrial fibrillation in patients with or without symptoms: the EAST-AFNET 4 trial. *Eur Heart J*. 2022;43(12):1219-1230. doi:10.1093/eurheartj/ehab593

78. de Denus S, Sanoski CA, Carlsson J, Opolski G, Spinler SA. Rate vs rhythm control in patients with atrial fibrillation: a meta-analysis. *Arch Intern Med*. 2005;165(3):258-262. doi:10.1001/archinte.165.3.258

79. Jaïs P, Cauchemez B, Macle L, et al. Catheter ablation versus antiarrhythmic drugs for atrial fibrillation: the A4 study. *Circulation*. 2008;118(24):2498-2505. doi:10.1161/CIRCULATIONAHA.108.772582

80. Andrade JG, Wells GA, Deyell MW, et al; EARLY-AF Investigators. Cryoablation or drug therapy for initial treatment of atrial fibrillation. *N Engl J Med*. 2021;384(4):305-315. doi:10.1056/NEJMoa2029980

81. Piccini JP, Lopes RD, Kong MH, Hasselblad V, Jackson K, Al-Khatib SM. Pulmonary vein isolation for the maintenance of sinus rhythm in patients with atrial fibrillation: a meta-analysis of randomized, controlled trials. *Circ Arrhythm Electrophysiol*. 2009;2(6):626-633. doi:10.1161/CIRCEP.109.856633

82. Voskoboinik A, Moskovitch JT, Harel N, Sanders P, Kistler PM, Kalman JM. Revisiting pulmonary vein isolation alone for persistent atrial fibrillation: a systematic review and meta-analysis. *Heart Rhythm*. 2017;14(5):661-667. doi:10.1016/j.hrthm.2017.01.003

83. Packer DL, Mark DB, Robb RA, et al; CABANA Investigators. Effect of catheter ablation vs antiarrhythmic drug therapy on mortality, stroke, bleeding, and cardiac arrest among patients with atrial fibrillation: the CABANA randomized clinical trial. *JAMA*. 2019;321(13):1261-1274. doi:10.1001/jama.2019.0693

84. Mark DB, Anstrom KJ, Sheng S, et al; CABANA Investigators. Effect of catheter ablation vs medical therapy on quality of life among patients with atrial fibrillation: the CABANA randomized clinical trial. *JAMA*. 2019;321(13):1275-1285. doi:10.1001/jama.2019.0692

85. Bahnson TD, Giczewska A, Mark DB, et al; CABANA Investigators. Association between age and outcomes of catheter ablation versus medical therapy for atrial fibrillation: results from the CABANA trial. *Circulation*. 2022;145(11):796-804. doi:10.1161/CIRCULATIONAHA.121.055297

86. Wazni OM, Marrouche NF, Martin DO, et al. Radiofrequency ablation vs antiarrhythmic drugs as first-line treatment of symptomatic atrial fibrillation: a randomized trial. *JAMA*. 2005;293(21):2634-2640. doi:10.1001/jama.293.21.2634

87. Oral H, Pappone C, Chugh A, et al. Circumferential pulmonary-vein ablation for chronic atrial fibrillation. *N Engl J Med*. 2006;354(9):934-941. doi:10.1056/NEJMoa050955

88. Khan MN, Jaïs P, Cummings J, et al; PABA-CHF Investigators. Pulmonary-vein isolation for atrial fibrillation in patients with heart failure. *N Engl J Med*. 2008;359(17):1778-1785. doi:10.1056/NEJMoa0708234

89. Wilber DJ, Pappone C, Neuzil P, et al; ThermoCool AF Trial Investigators. Comparison of antiarrhythmic drug therapy and radiofrequency catheter ablation in patients with paroxysmal atrial fibrillation: a randomized controlled trial. *JAMA*. 2010;303(4):333-340. doi:10.1001/jama.2009.2029

90. Cosedis Nielsen J, Johannessen A, Raatikainen P, et al. Radiofrequency ablation as initial therapy in paroxysmal atrial fibrillation. *N Engl J Med*. 2012;367(17):1587-1595. doi:10.1056/NEJMoa1113566

91. Packer DL, Kowal RC, Wheelan KR, et al; STOP AF Cryoablation Investigators. Cryoballoon ablation of pulmonary veins for paroxysmal atrial fibrillation: first results of the North American Arctic Front (STOP AF) pivotal trial. *J Am Coll Cardiol*. 2013;61(16):1713-1723. doi:10.1016/j.jacc.2012.11.064

92. Morillo CA, Verma A, Connolly SJ, et al; RAAFT-2 Investigators. Radiofrequency ablation vs antiarrhythmic drugs as first-line treatment of paroxysmal atrial fibrillation (RAAFT-2): a randomized trial. *JAMA*. 2014;311(7):692-700. doi:10.1001/jama.2014.467

93. Di Biase L, Mohanty P, Mohanty S, et al. Ablation versus amiodarone for treatment of persistent atrial fibrillation in patients with congestive heart failure and an implanted device: results from the AATAC multicenter randomized trial. *Circulation*. 2016;133(17):1637-1644. doi:10.1161/CIRCULATIONAHA.115.019406

94. Marrouche NF, Brachmann J, Andresen D, et al; CASTLE-AF Investigators. Catheter ablation for atrial fibrillation with heart failure. *N Engl J Med*. 2018;378(5):417-427. doi:10.1056/NEJMoa1707855

95. Roy D, Talajic M, Nattel S, et al; Atrial Fibrillation and Congestive Heart Failure Investigators. Rhythm control versus rate control for atrial fibrillation and heart failure. *N Engl J Med*. 2008;358(25):2667-2677. doi:10.1056/NEJMoa0708789

96. Torp-Pedersen C, Møller M, Bloch-Thomsen PE, et al; Danish Investigations of Arrhythmia and Mortality on Dofetilide Study Group. Dofetilide in patients with congestive heart failure and left ventricular dysfunction. *N Engl J Med*. 1999;341(12):857-865. doi:10.1056/NEJM199909163411201

97. Hunter RJ, Berriman TJ, Diab I, et al. A randomized controlled trial of catheter ablation versus medical treatment of atrial fibrillation in heart failure (the CAMTAF trial). *Circ Arrhythm Electrophysiol*. 2014;7(1):31-38. doi:10.1161/CIRCEP.113.000806

© 2024 American Medical Association. All rights reserved, including those for text and data mining, AI training, and similar technologies.

Downloaded from jamanetwork.com by Northern Arizona University user on 05/25/2025

98. Hsu LF, Jaïs P, Sanders P, et al. Catheter ablation for atrial fibrillation in congestive heart failure. *N Engl J Med*. 2004;351(23):2373-2383. doi:10.1056/NEJMoa041018

99. Turagam MK, Garg J, Whang W, et al. Catheter ablation of atrial fibrillation in patients with heart failure: a meta-analysis of randomized controlled trials. *Ann Intern Med*. 2019;170(1):41-50. doi:10.7326/M18-0992

100. Al Halabi S, Qintar M, Hussein A, et al. Catheter ablation for atrial fibrillation in heart failure patients: a meta-analysis of randomized controlled trials. *JACC Clin Electrophysiol*. 2015;1(3):200-209. doi:10.1016/j.jacep.2015.02.018

101. Simader FA, Howard JP, Ahmad Y, et al. Catheter ablation improves cardiovascular outcomes in patients with atrial fibrillation and heart failure: a meta-analysis of randomized controlled trials. *Europace*. 2023;25(2):341-350. doi:10.1093/europace/euac173

102. Packer DL, Piccini JP, Monahan KH, et al; CABANA Investigators. Ablation versus drug therapy for atrial fibrillation in heart failure: results from the CABANA trial. *Circulation*. 2021;143(14):1377-1390. doi:10.1161/CIRCULATIONAHA.120.050991

103. Benjamin EJ, Thomas KL, Go AS, et al. Transforming atrial fibrillation research to integrate social determinants of health: a National Heart, Lung, and Blood Institute workshop report. *JAMA Cardiol*. 2023;8(2):182-191. doi:10.1001/jamacardio.2022.4091

104. Abdel-Qadir H, Akioyamen LE, Fang J, et al. Association of neighborhood-level material deprivation with atrial fibrillation care in a single-payer health care system: a population-based cohort study. *Circulation*. 2022;146(3):159-171. doi:10.1161/CIRCULATIONAHA.122.058949

105. The Joint Commission. R3 report issue 36: new requirements to reduce health care disparities. June 20, 2022. Accessed October 10, 2024. https://www.jointcommission.org/standards/r3-report/r3-report-issue-36-new-requirements-to-reduce-health-care-disparities/

106. Heilman E. An intro to CMS's SDOH measures. Medisolv. October 7, 2022. Accessed October 10, 2024. https://blog.medisolv.com/articles/intro-cms-sdoh-measures

107. Sarraju A, Maron DJ, Rodriguez F. Under-reporting and under-representation of racial/ethnic minorities in major atrial fibrillation clinical trials. *JACC Clin Electrophysiol*. 2020;6(6):739-741. doi:10.1016/j.jacep.2020.03.001

© 2024 American Medical Association. All rights reserved, including those for text and data mining, AI training, and similar technologies.

Downloaded from jamanetwork.com by Northern Arizona University user on 05/25/2025